

Toronto Homicide Patterns Analysis

(2004–2025)

BDA800: Business Analytics | Project Milestone 4 | Group 7

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1. Abstract

This report looks at homicide cases in Toronto from 2004 to 2025 using data from the Toronto Police. It follows a step-by-step data analysis approach and focuses on three main things: understanding patterns in the data, building models to predict outcomes, and suggesting ways police can improve how they use their resources.

The dataset includes 1,526 cases over 21 years. From the analysis, it's clear that most homicides involve shootings (about half of all cases). The crimes are also not spread evenly across the city; they tend to happen more in certain neighbourhoods and police divisions. In terms of timing, incidents are more common in July and August, and weekends tend to have higher numbers as well.

Two prediction models were tested and compared with a simple baseline. The Decision Tree didn't perform very well, but the Random Forest model did slightly better and handled the complex patterns in the data more effectively.

Based on these results, the report suggests focusing police efforts more on high-crime divisions like 31, 51, and 42, increasing safety measures during the summer, putting more effort into preventing gun-related crimes, and using the Random Forest model to help make better real-time decisions.

2. Introduction and Problem Definition

Public safety is a major concern in any big city, and Toronto is no different. Even though it's considered one of the safest large cities in Canada, homicide cases still happen and have shown noticeable increases in certain years, like 2005 and 2018. To deal with this effectively, it's important to understand when these incidents happen, where they occur, and what types are most common so that police can respond better and prevent future cases.

This project was carried out to provide clear, data-based insights into homicide patterns in Toronto. It is mainly useful for Toronto Police leaders, government decision-makers, and public safety planners who need practical information to guide their actions. The aim is not just to study past trends, but also to build a system that helps predict future risks, allowing authorities to take action early instead of only reacting after incidents occur.

2.1 Problem Statement

Homicide cases in Toronto are largely clustered in certain neighbourhoods and are mainly driven by shootings, which consistently occur at significantly higher rates than other homicide types over time. The challenge is threefold:

- Identifying geographic hotspots that receive disproportionately high crime activity.
- Understanding temporal patterns that correlate with elevated incident rates.
- Classifying the type of homicide (Shooting, Stabbing, or Other) using geographic and temporal signals to enable better emergency resource dispatch.

2.2 Analytical Framework: CRISP-DM

The project used the CRISP-DM method, which is a clear step-by-step approach for doing data analysis. It helped guide the entire process, from understanding the problem, preparing the data, building models, and evaluating results, to planning how the solution can be used in real life.

Using this method made sure that every step of the analysis stayed focused on the actual needs of the Toronto Police and city planners, so the results are practical and useful, not just theoretical.

3. Dataset and Research Questions

3.1 Dataset Overview

The primary dataset is the Toronto Police Service Homicide Open Data, accessed through the City of Toronto Open Data Portal. It contains records of all homicide incidents reported to TPS between 2004 and 2025.

Attribute	Detail
Total Incidents	1,526 records
Features	18 original columns
Time Span	2004 – 2025 (21 years)
Homicide Types	Shooting, Stabbing, Other
Geographic Scope	City of Toronto (latitude/longitude + neighbourhood)
Missing Values	0 (post-cleaning)
Duplicate Records	0 (post-cleaning)

Two supplementary sources were used for context and validation: the City of Toronto's 158-Neighbourhood Boundary dataset (to support geographic analysis), and Statistics Canada's Homicide Survey (Survey 3315), which provided national-level benchmarks against which Toronto's trends could be compared.

3.2 Research Questions

The project was guided by five core research questions:

- Which neighbourhoods in Toronto have the highest and lowest numbers of homicide incidents?
- What type of homicide occurs most frequently, and how has this distribution changed over time?
- Is there a specific month or day of the week when homicide incidents occur most frequently?
- Which year had the highest homicide rate in Toronto?
- Are homicide incidents geographically clustered in certain areas of Toronto, and do these clusters correspond to specific police divisions?

4. Data Processing and Cleaning

The raw data from the Toronto Police open portal couldn't be used directly for machine learning, so it first had to be cleaned and prepared. This preprocessing was done in three main stages to make sure the data was accurate, consistent, and ready for analysis.

4.1 Data Cleaning

- No missing values were identified in the final dataset.
- No duplicate records were present.
- The OCC_DATE field was converted from string to datetime format to enable temporal feature extraction.
- All latitude and longitude coordinates were validated to fall within the geographic bounds of the City of Toronto.

4.2 Column Removal

Several features were dropped to reduce dimensionality and remove redundancy:

- OBJECTID and EVENT_UNIQUE_ID: internal identifiers with no predictive value.
- HOOD_140 and NEIGHBOURHOOD_140: legacy neighbourhood classification scheme, superseded by the 158-neighbourhood system.
- Projected x/y coordinate columns, replaced by WGS84 latitude and longitude for geographic consistency.

4.3 Feature Engineering

Seven predictive input features were selected or engineered for the classification models:

Feature	Description	Encoding
DIVISON_ENC	Police division	Label-encoded
NEIGHBOURHOOD_ENC	Neighbourhood (158-system)	Label-encoded
OCC_YEAR	Year of occurrence	Numeric
OCC_MONTH	Month of occurrence	Label-encoded
OCC_DOW	Day of week	Label-encoded
LAT_WGS84	Latitude coordinate	Numeric
LONG_WGS84	Longitude coordinate	Numeric

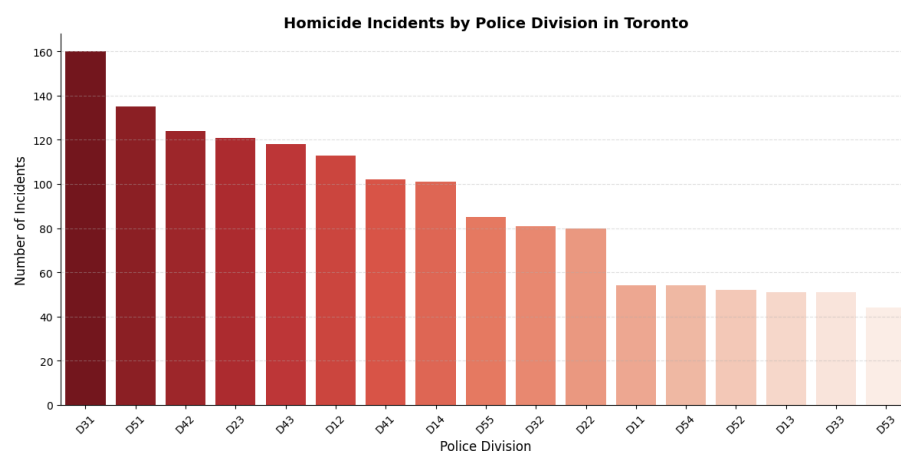
The target variable, HOMICIDE_TYPE, was encoded as a three-class label (Shooting = 1, Stabbing = 2, Other = 0). An 80/20 stratified train-test split (random_state=42) was applied, yielding 1,220 training records and 306 test records.

5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was done first to better understand the data before building any models. This step helped identify key patterns, unusual values, and relationships within the dataset. The analysis focused on four main areas: where incidents happened across the city, the different types of homicide, how cases changed over time, and how specific types of homicide varied across different time periods.

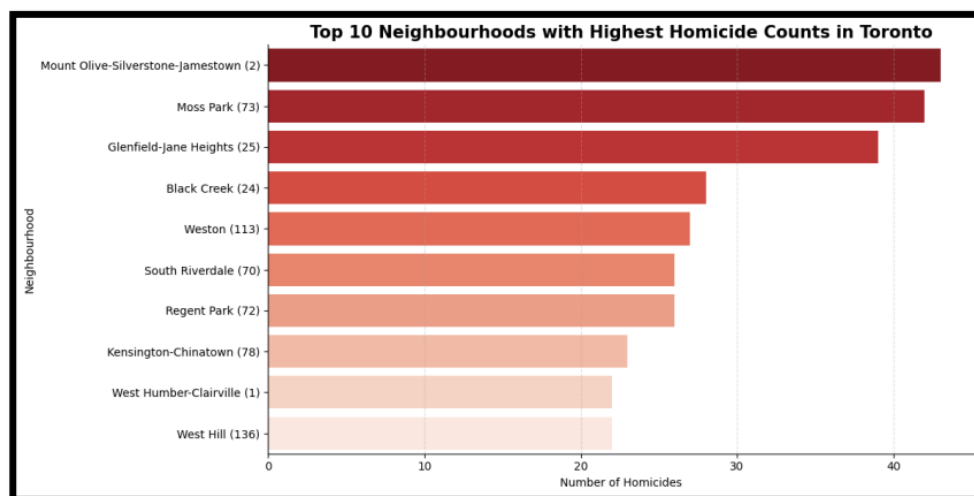
5.1 Geographic Hotspots

Homicide cases in Toronto are not spread evenly across all 158 neighbourhoods. Instead, a small number of areas experience a much higher number of incidents compared to others.



The three main police divisions with the highest concentration of cases are Division 31 (North York/West End), Division 51 (Downtown East), and Division 42 (Scarborough/North East). These areas account for a large share of the city's total incidents, making them key focus points for policing and prevention efforts.

Top Toronto Neighbourhoods with the Highest Homicide Concentration



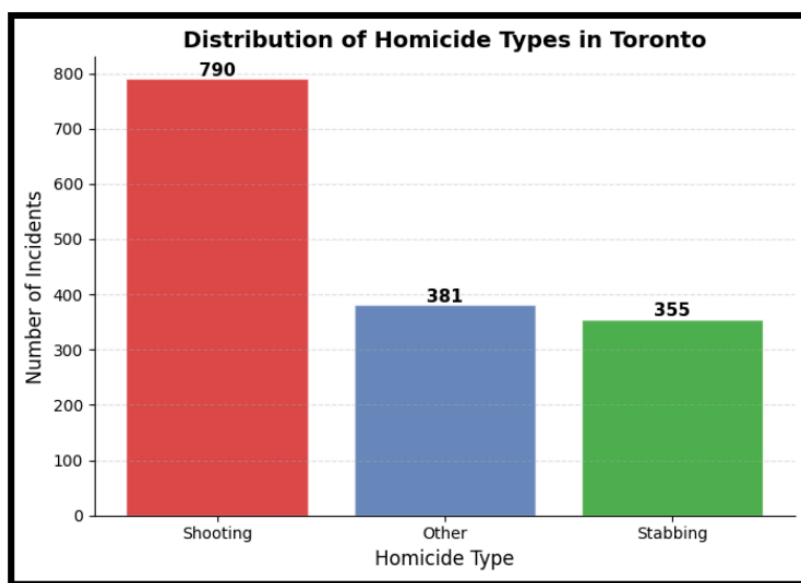
Mount Olive-Silverstone-Jamestown has the highest number of cases, with more than 44 incidents over the study period. It is followed by Moss Park and Glenfield-Jane Heights.

This shows that homicides are heavily concentrated in specific areas, suggesting that focused, neighbourhood-level strategies would be more effective than applying the same approach across the entire city.

5.2 Homicide Type Distribution

Shooting is the dominant homicide type with 790 incidents (approximately 52% of the dataset), more than double the frequency of stabbing (355) or other methods (381). This finding aligns with national trends reported by Statistics Canada and highlights the central role of gun violence in Toronto's homicide landscape.

Distribution of Homicide Types Shows Shooting as the Leading Cause

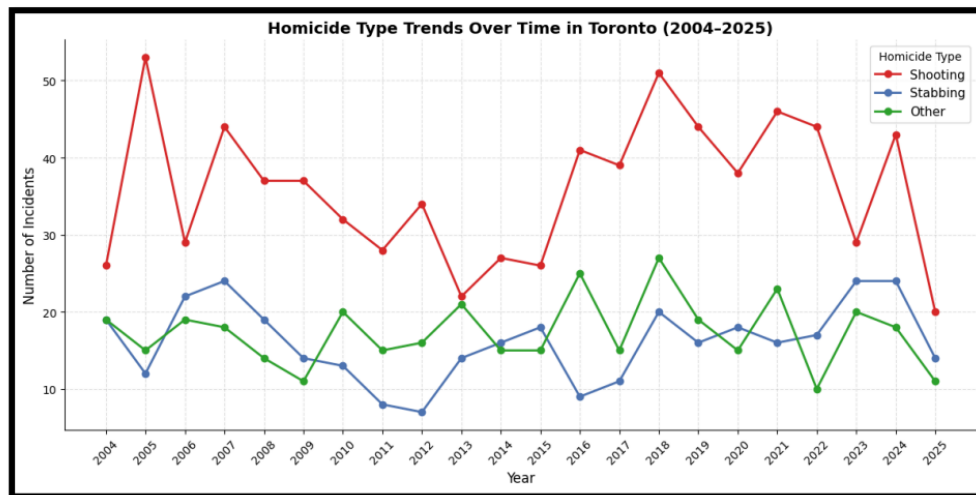


5.3 Temporal Patterns

The time-based analysis showed two clear and consistent patterns. First, at the monthly level, homicide cases increase noticeably in July and August, which is likely due to more outdoor activities and social interactions during the summer.

Second, when looking at days of the week, incidents are higher from Friday to Sunday compared to weekdays. These patterns suggest that police can be more effective by increasing patrols and resources during these high-risk times.

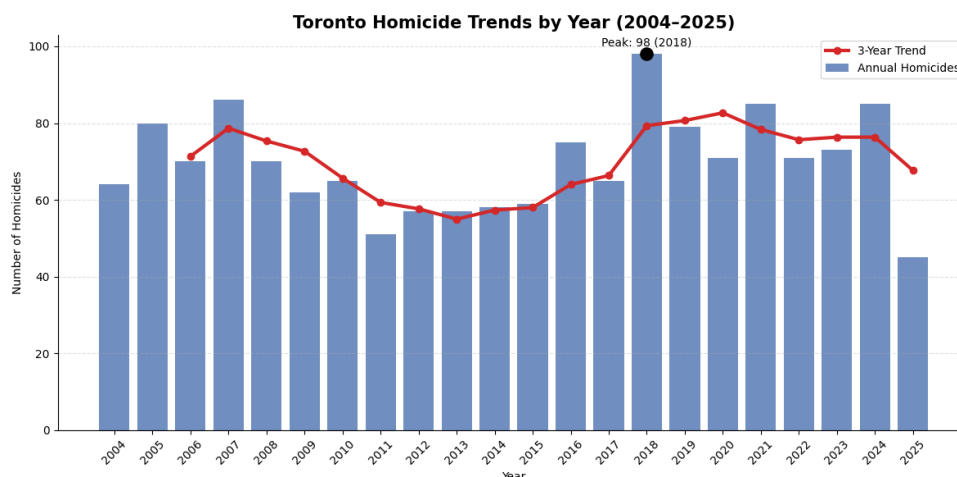
Shooting Homicides Consistently Occur at Higher Rates Over Time in Toronto



5.4 Homicide Type Trends Over Time (2004–2025)

It shows how different types of homicides have changed over time. Shootings are clearly the most common type every year, with major spikes in 2005 and 2018. Since around 2016, shooting incidents have stayed consistently high, which highlights the need for continued focus on preventing gun-related violence.

In comparison, stabbing and other types of homicide have remained relatively steady over the years, without any major long-term increase.



6. Implementation and Code

Two data mining algorithms were implemented for this project: a Decision Tree classifier and a Random Forest ensemble classifier. Both models were built using Python's scikit-learn library, following the feature engineering and train-test split.

6.1 Baseline Model: Zero Rule

Before training any machine learning model, a zero-rule baseline was established. This classifier predicts the most frequent class ('Shooting', Class 1) for every instance in the test set. The Zero Rule achieved 54.25% accuracy on the 306-record test set, representing the performance floor that any useful model must exceed. This baseline is critical: a model failing to surpass 54.25% provides no genuine predictive value over naive guessing.

6.2 Model 1: Decision Tree Classifier

Configuration: DecisionTreeClassifier(max_depth=5, random_state=42) with 7 input features.

The Decision Tree was configured with a maximum depth of 5 to prevent overfitting on the training data. Key results:

- Test Accuracy: 51.63%
- 5-Fold Cross-Validation Mean: 48.89% | CV Std: 0.0125
- The model demonstrated stable generalization, the narrow gap between CV and test accuracy confirmed it was not memorizing training data.
- However, it failed to surpass the Zero Rule baseline (54.25%), largely due to severe class imbalance bias toward 'Shooting'.
- Minority classes (Stabbing, Other) were heavily misclassified, pointing to a need for class-balancing techniques or richer features to improve minority-class precision.

6.3 Model 2: Random Forest Classifier

Configuration: RandomForestClassifier (n_estimators=200, random_state=42) with 7 input features.

The Random Forest ensemble, comprising 200 decision trees, substantially outperformed the single Decision Tree. Key results:

- Test Accuracy: 56.21% (surpasses the 54.25% baseline)
- 5-Fold Cross-Validation Mean: 52.29% | CV Std: 0.0159
- Shooting class recall: 83% (137 correct predictions out of 166 test instances).
- Other/Stabbing precision improved to ~38%, reflecting better bias-variance trade-off vs. the Decision Tree.
- The 4% gap between CV and test accuracy suggests some sensitivity to the specific composition of the 20% test slice, but the low standard deviation confirms overall robustness.

The ensemble approach leveraged 200 trees to capture complex, non-linear interactions between geographic and temporal features, interactions that a single depth-5 tree cannot adequately model.

7. Results, Ideas, and Insights

7.1 Model Comparison Summary

Model	Test Accuracy	CV Mean (5-Fold)	CV Std	vs. Baseline
Baseline (Zero Rule)	54.25%	N/A	N/A	--
Model 1: Decision Tree	51.63%	48.89%	0.0125	-2.62%
Model 2: Random Forest	56.21%	52.29%	0.0159	+1.96%

The Random Forest model performed the best among all the models tested. Even though its accuracy is 56.21%, which might seem low at first, it actually makes sense given the situation. The model is trying to classify three different types of homicide using a relatively small dataset of 1,526 cases, and the data is unevenly distributed (with shootings making up over half the cases). This makes the task quite challenging.

Despite this, the model still performs better than simple guessing and is able to pick up useful patterns, especially for predicting shooting incidents, which are the most common type.

7.2 Confusion Matrix Analysis

Analysis of the Random Forest confusion matrix reveals three distinct performance profiles by class:

- Shooting (Strong): 137 correct out of 166 test instances (83% recall). The model excels here because Shooting has the most training examples and the clearest class boundary in feature space.
- Other (Weak): High confusion with other classes, with significant mispredictions into Shooting. The 'Other' category is heterogeneous by nature, lacking consistent geographic or temporal signals.
- Stabbing (Weak): The most challenging class, only 13 correct predictions out of 64 instances (20% recall). The majority are misclassified as Shooting or Other, indicating severe feature overlap between these categories.

These results suggest that additional features, such as victim demographics, suspect demographics, or scene characteristics, could substantially improve minority-class performance in future iterations of the model.

7.3 Parameter Tuning: Decision Tree max_depth

To determine the optimal tree depth for the Decision Tree classifier, we tested max_depth values ranging from 1 to 20 and recorded the 5-fold cross-validated accuracy at each setting. The results, summarised in Table 2 below, show how model performance changes as the tree are allowed to grow deeper.

The highest cross-validated accuracy was observed at max_depth = 1 (51.77%), which is marginally below the Zero Rule baseline of 54.25%. Performance then declines and stabilises as depth increases, confirming that the Decision Tree overfits when given more freedom. This result demonstrates that homicide type classification is too complex for simple rule-based splits and directly motivated our choice of the Random Forest ensemble as the primary model.

Table 2: Decision Tree CV Accuracy by max_depth (selected values)

max_depth	CV Accuracy (%)	Note
1	51.77	Best depth
3	50.42	
5	48.89	Final model config
10	49.11	Overfitting begins
20	48.30	Performance plateaus

7.4 Key Insights

- The most important and practical insight is the geographic concentration of crime: just three divisions (D31, D51, D42) and a few neighbourhoods are responsible for a large share of homicides over the 21 years.
- Clear time patterns can be used effectively: the consistent rise in cases during July-August and on weekends makes it possible to plan patrol increases.
- The rise in gun-related violence since 2016 is not temporary; it's a long-term trend that requires ongoing and focused prevention efforts.
- The Random Forest model performs better than a single Decision Tree, proving that using an ensemble approach is more suitable for this type of complex data and supporting its use in real-world deployment.

8. Prescriptive Recommendations

Based on the combined findings from EDA, descriptive analysis, and predictive modelling, four actionable recommendations are proposed for the Toronto Police Service and city policymakers.

8.1 Hyper-Local Resource Allocation

Recommendation:

Increase community policing efforts in Divisions 31 (North York/West End), 51 (Downtown East), and 42 (Scarborough/North East). This should include deploying dedicated units in high-risk areas like Mount Olive-Silverstone-Jamestown.

Data-Driven Basis:

Analysis of 21 years of data consistently shows that these locations have a much higher concentration of homicide incidents. This means broad, city-wide patrol

strategies are less effective. Instead, focusing resources in these specific hotspots can lead to better outcomes and more efficient use of time and funding.

8.2 Summer Safety Surge (July-August)

Recommendation:

Implement a proactive “Safety Surge” during the third quarter (July-August) each year. This should include increased police presence and stronger community engagement, especially during weekend evenings when risk is highest.

Data-Driven Basis:

The time-based analysis consistently shows that these months have the highest number of incidents across the entire 2004-2025 period. Since this pattern is stable and predictable, planning resources is both practical and cost-effective, allowing authorities to act before incidents occur rather than responding afterward.

8.3 Firearm-Specific Prevention Programs

Recommendation:

Redirect a significant portion of public safety resources toward firearm-focused initiatives. This includes programs like gun buybacks, stronger enforcement against illegal firearms, and youth diversion efforts aimed at preventing involvement in gun violence, rather than relying mainly on general anti-violence programs.

Data-Driven Basis:

The analysis shows that shootings make up about 52% of all homicide cases and have been the main reason for the increase in incidents since 2016. In contrast, other types of homicide, like stabbings, have stayed relatively stable. This imbalance suggests that focusing more resources on gun-related issues will have a greater impact on reducing overall homicide rates.

8.4 Deploy Random Forest as a Decision-Support Tool

Recommendation:

Use the Random Forest model in TPS dispatch systems so that, when an incident is reported, it can quickly estimate the likely type of homicide in real time.

Data-Driven Basis:

The model achieves about 56.21% accuracy using only basic information like location and time, which is better than random guessing. Even a small improvement like this can be very useful in real situations. For example, if an incident is reported in a high-risk area on a summer weekend night, the model may indicate a higher chance of it being a shooting. This allows dispatchers to send the right type of response team—such as specialized firearms units—instead of general officers, helping reduce response time and handle the situation more effectively.

9. Conclusion

This project successfully used a full CRISP-DM approach to analyse 21 years of homicide data from the Toronto Police Service. Starting with raw open data, the team cleaned and prepared the dataset, carried out detailed analysis across

location, time, and types of incidents, and built two machine learning models that were compared against a simple baseline.

The Random Forest model performed the best, with an accuracy of 56.21%, slightly higher than the baseline of 54.25%. It works well because it can capture complex relationships between factors like location, neighbourhood, police division, and timing—insights that are directly useful for real-world policing decisions.

In addition to the model, the analysis itself provided valuable insights. It clearly identified high-risk areas, showed consistent spikes in July and August, highlighted the growing impact of gun violence since 2016, and revealed that a large portion of incidents are concentrated in just three police divisions. These findings offer a strong foundation for smarter, data-driven resource planning.

For future improvements, the project could include more detailed data, such as socioeconomic factors, population density, or nearby past incidents. It could also handle data imbalance better using techniques like SMOTE or weighted training, and explore additional methods like clustering (e.g., K-Means or DBSCAN) to better identify and understand crime hotspots.

10. Reflection

Working on this project was a valuable and rewarding experience that deepened our understanding of applied data analytics. The most straightforward part of the project was the exploratory data analysis: once the dataset was cleaned, visualizing geographic hotspots and temporal patterns was intuitive and produced clear, meaningful results. Seeing the city-wide clustering of homicides in specific divisions and the consistent summer spikes confirmed that real-world data can tell compelling stories when approached methodically.

The most challenging aspect was dealing with class imbalance in the classification task. Despite trying multiple configurations, the models consistently struggled to predict minority classes (Stabbing and Other) with high recall. This taught us an important lesson about the limits of standard classifiers on imbalanced datasets and motivated us to explore potential remedies such as SMOTE and class-weighted training for future iterations. Understanding why accuracy alone is not a sufficient evaluation metric and learning to interpret precision, recall, and confusion matrices holistically were two of the most instructive parts of the project.

The most enjoyable part of the project was translating analytical findings into practical, policy-relevant recommendations. Connecting data patterns to concrete actions, such as deploying the Random Forest model in dispatch systems or designing a summer safety surge program, made the work feel genuinely impactful.

11. References

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Declaration and Contribution

We, Karishma Nainwani, Naila Farooqi, and Togashan Pirapakaran, declare that the attached milestone is our own work in accordance with the Seneca Academic Policy. We have not copied any part of this assignment, manually or electronically, from any other source, including websites, unless specified as references. We have not distributed our work to other students.

Member Contributions:

1. Karishma Nainwani: Data preparation & cleaning, feature engineering, writing report
2. Naila Farooqi: EDA, visualizations, model implementation, parameter tuning, writing report, Slides
3. Togashan Pirapakaran: Research, model evaluation, conclusion, writing report, Slides